

Review: The Sequel

General Topics

Probability

- Specific Rules
- Multiple Events
- Expected Value

Distributions

- Binomial Distribution
- Sampling Distributions
- Law of Large Numbers
- Central Limit Theorem

Hypothesis Testing

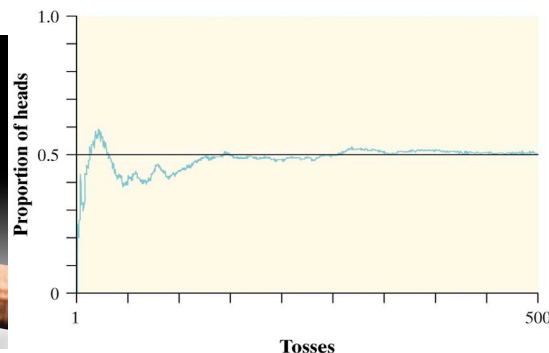
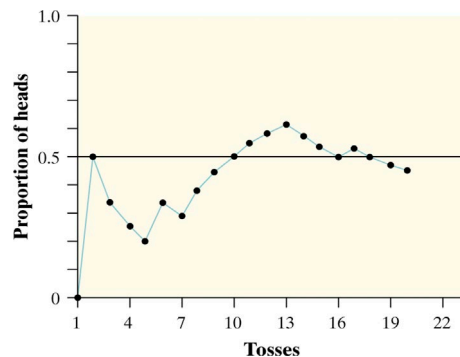
- t-tests
- ANOVA

Causal Inference

Probability

Probability Definitions

- Big Idea: Chance behavior is unpredictable in the short run, but it has a regular and predictable pattern in the long run.
- Simple def. Way to quantify uncertainty
- Textbook def. Mathematics of chance behavior
- Operational def. The probability of any outcome of a random phenomenon is the proportion of times the outcome would occur in a very long series of repetitions.
 - A phenomenon is random if individual outcomes are uncertain but there is, nonetheless, a regular distribution of outcomes in a large number of repetitions



More Definitions

- Event – outcome or set of outcomes of a random phenomenon
 - an event is a subset of the sample space
- Sample space – set of all possible outcomes
 - Example : coin flip
 - Possible outcomes: heads or tails
 - $\{H, T\}$
 - Die roll
 - $\{1, 2, 3, 4, 5, 6\}$
 - Cards in a deck
 - $\{\text{Ace of clubs... 2 of clubs, ..., A hearts, A spades, etc}\}$



Even More Definitions

Probability model

- a mathematical description of a random phenomenon consisting of two parts:
 - a sample space S and
 - a way of assigning probabilities to events

Classical def of probability

- A is a set of outcomes in sample space S
- $P(A) = \# \text{ of events in } A / \# \text{ of events in } S$
- Example
 - $P(\text{king}) = 4/52 = 1/13$
 - $P(\text{king of spades}) = 1/52$

Properties of Probabilities

Any probability is a number between 0 and 1.

- $0 \leq P(A) \leq 1$

All possible outcomes together must have probability 1.

- Because some outcome must occur on every trial,
 - the sum of the probabilities for all possible outcomes must be exactly 1
- $P(S) = 1$
 - Probability of sample space must be 1

Properties of Probabilities

- If two events have no outcomes in common (disjoint), the probability that one or the other occurs is the sum of their individual probabilities.
 - $P(A \text{ or } B) = P(A) + P(B)$ IF disjoint
 - Disjoint (A and B don't overlap)
- The probability that an event does not occur is 1 minus the probability that the event does occur.
 - The probability that an event occurs and the probability that it does not occur always add to 1, or 100%.
 - $P(\text{not } A) = 1 - P(A)$

Abridged Properties of Probabilities

- $0 \leq P(A) \leq 1$
- $P(S) = 1$
 - Probability of sample space must be 1
- $P(A \text{ or } B) = P(A) + P(B)$ IF disjoint
 - Disjoint (A and B don't overlap)
- $P(\text{not } A) = 1 - P(A)$

Finite probability models

- One way to assign probabilities to events is to assign a probability to every individual outcome, then add these probabilities to find the probability of any event. This idea works well when there are only a finite (fixed and limited) number of outcomes.
- A probability model with a finite sample space is called a finite probability model.

Finite probability models

- To assign probabilities in a finite model, list the probabilities of all the individual outcomes. These probabilities must be numbers between 0 and 1 that add to exactly 1. The probability of any event is the sum of the probabilities of the outcomes making up the event.
- Finite probability models are sometimes called discrete probability models. Statisticians often refer to finite probability models as discrete.

Continuous probability models

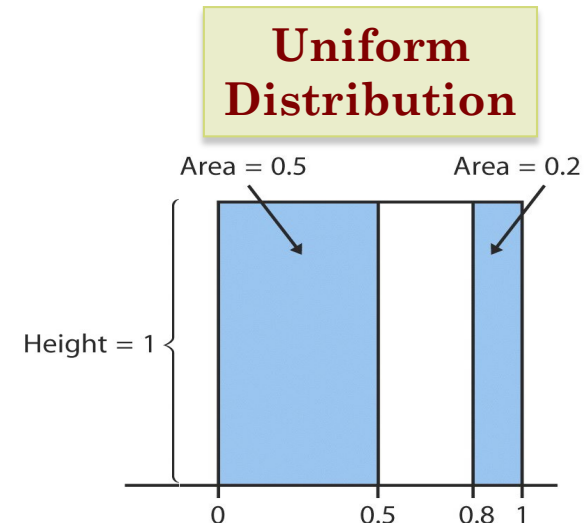
Suppose we want to choose a number at random between 0 and 1, allowing any number between 0 and 1 as the outcome.

We cannot assign probabilities to each individual value because there is an infinite continuum of possible values.

A continuous probability model assigns probability as an area under a density curve. The area under the curve and above any range of values is the probability of an outcome in that range.

Example: Find the probability of getting a random number that is less than or equal to 0.5 or greater than 0.8.

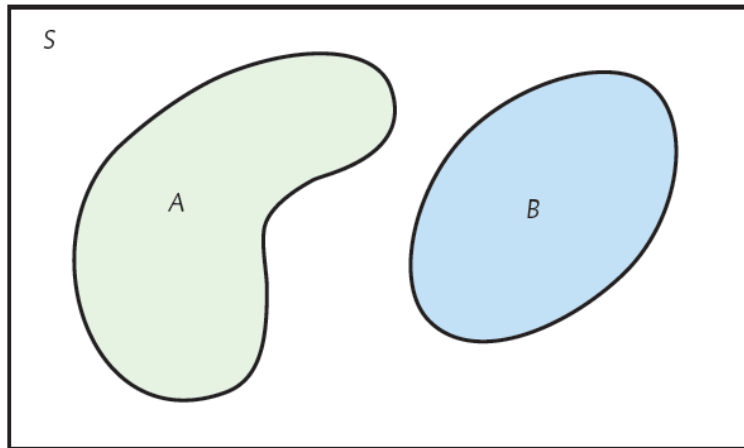
$$\begin{aligned} &P(X \leq 0.5 \text{ or } X > 0.8) \\ &= P(X \leq 0.5) + P(X > 0.8) \\ &= 0.5 + 0.2 = 0.7 \end{aligned}$$



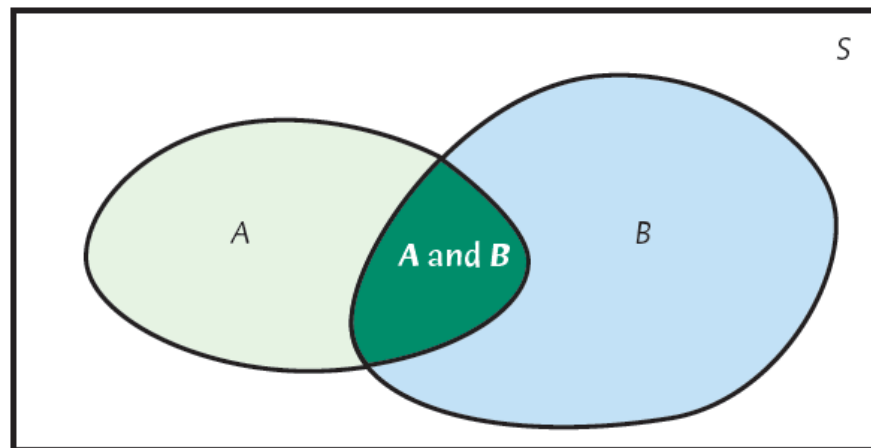
Multiple Events

- The upcoming rules look at multiple events.
- I find it helpful to draw a picture to display relations among several events. A picture that shows the sample space S as a rectangular area and shows events as areas within S is called a Venn diagram.

Two events that are disjoint.



Two events that are not disjoint.
The event $\{A \text{ and } B\}$ consists of the outcomes that they have in common.



The general addition rule

We know that if A and B are disjoint events:

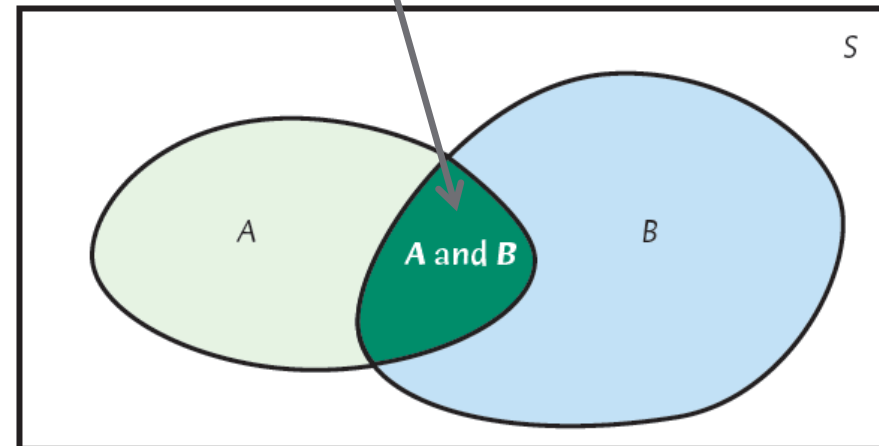
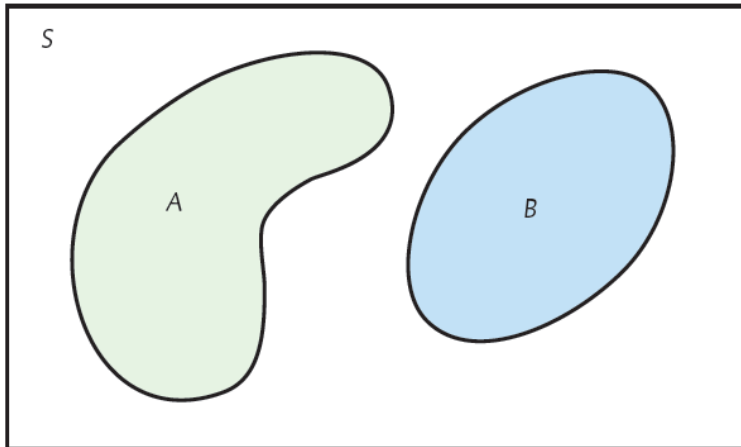
$$P(A \text{ or } B) = P(A) + P(B)$$

Addition rule for *any* two events

For any two events A and B:

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

Outcomes here are double-counted by $P(A) + P(B)$.



The general multiplication rule

The probability that two events A and B happen together can be found by

$$P(A \text{ and } B) = P(A)P(B | A)$$

Here $P(B | A)$ is the conditional probability that B occurs, given the information that A occurs.

This may be extended to any number of events:

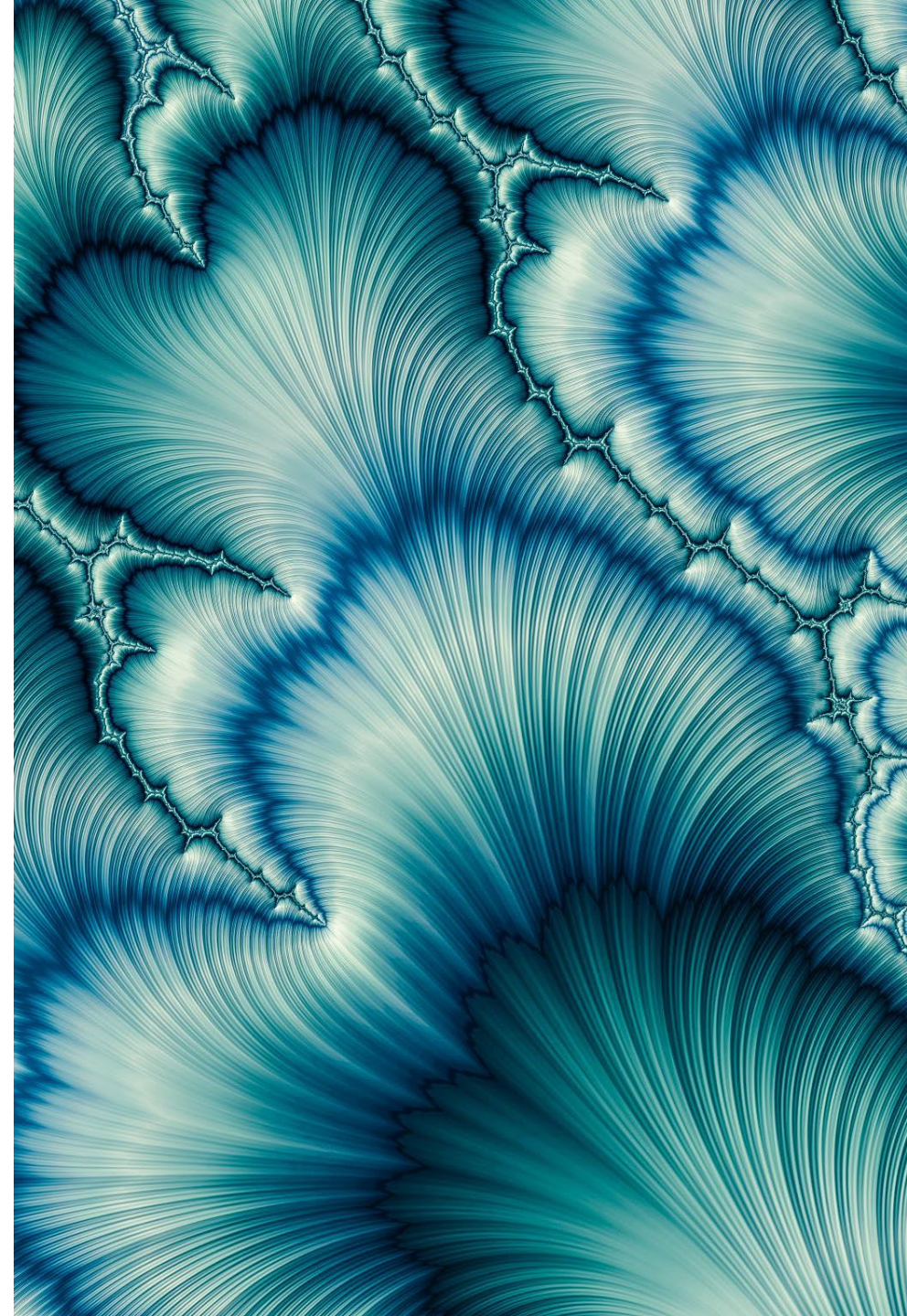
$$\begin{aligned} &P(A \text{ and } B \text{ and } C) \\ &= P(A)P(B | A)P(C | \text{both } A \text{ and } B) \end{aligned}$$

Conditional Probabilities Formalized

- The probability we assign to an event can change if we know that some other event has occurred.
 - If the occurrence of event B, depends in some sense on the occurrence of event A,
 - then we can talk about the conditional probability of B, given A.
- $P(B|A)$
 - B given A
- When $P(A) > 0$, the conditional probability of B given A is:

$$P(B | A) = \frac{P(A \text{ and } B)}{P(A)}$$

Visual Example: <http://setosa.io/conditional/>



Expected Values

- Expected value is exactly what you might think it means intuitively: the **return you can expect for some kind of action**
- For the two dice games, I showed the expected value for 100 games.
- The basic expected value formula is the **probability of an event multiplied by the amount of times the event happens:** $(P(x) * n)$.



Expected Values

- For example, if you take a 20 question multiple-choice test with A,B,C,D as the answers, and you guess all “A”, then you can expect to get 25% right (5 out of 20).
- The **math** behind this kind of expected value is:
The probability (P) of getting a question right if you guess: .25
The number of questions on the test (n)*: 20
 $P \times n = .25 \times 20 = 5$



Expected Value for Multiple Events

- Of course, calculating expected value (EV) gets more complicated in real life.
- For example, You buy two \$10 raffle tickets for a new car valued at \$15,000.
- Two thousand tickets are sold. What is the EV of your gain? The formula for calculating the EV where there are **multiple probabilities** is:
$$E(X) = \sum(X * P(X))$$
- The expected value for your 2 \$10 raffle tickets are:
 - $\$15,000 * 1/2000 + \$15,000 * 1/2000 = \$15$
- Would you buy the tickets?
- At what ticket price would you be indifferent?

Bayes's rule

- Bayes's rule provides a unifying structure and is a basic result in an important area of statistics known as Bayesian statistics.
- Example: In the last example, we were given $P(\text{online dating} \mid 18 \text{ to } 24) = 0.27$ (if we chose a random 18- to 24-year-old, the probability that they had used an online dating site was 0.27). But what if we want $P(18 \text{ to } 24 \mid \text{online dating})$?

Bayes's rule

- Suppose $B_1, B_2, \dots, B_n$ are disjoint events with positive probabilities, and the sum of these probabilities is 1.
- If A is any event with probability greater than 0, then

$$P(B_i \mid A) = \frac{P(A \mid B_i)P(B_i)}{P(A \mid B_1)P(B_1) + P(A \mid B_2)P(B_2) + \cdots + P(A \mid B_n)P(B_n)}$$

2. What is the probability that both children are boys?

- Mr. Smith has two children. At least one of them is a boy. What is the probability that both children are boys?
- If we formulate the question as:
 - From all families with two children,
 - one child is selected at random,
 - and the sex of that child is specified to be a boy.
- Bayes Formula:
 - $P(BB | b) = P(b | BB) * P(BB) / P(b)$
 - $P(b | BB) = 1$
 - $P(BB) = 1/4$
 - $P(b) = \text{probability of a boy} = 1/2$
 - $1 * .25 / .5 = 1/2$

Tree diagrams

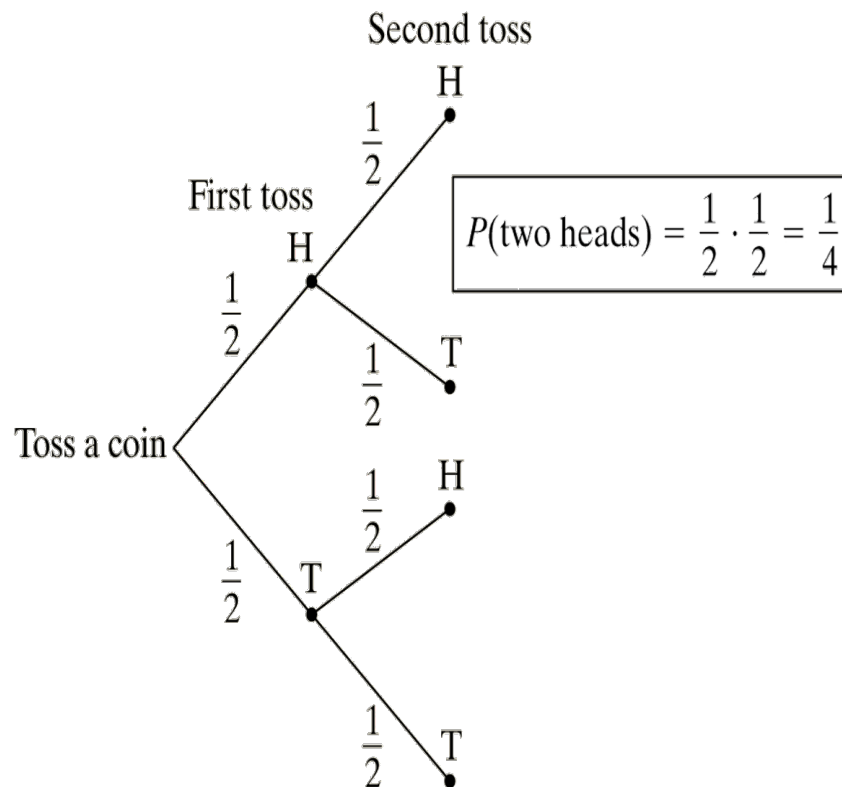
- We learned how to describe the sample space S of a chance process
- Another way to model chance behavior that involves a sequence of outcomes is to construct a tree diagram.

Consider flipping a coin twice. What is the probability of getting two heads?

Sample Space

HH HT TH TT

So, $P(\text{two heads}) = P(HH) = 1/4$.



Distributions

Binomial Distributions

- Just like the normal distribution describes bell-shaped data, the binomial distribution describes count data.
- Specifically, it models the number of successful outcomes,
 - when the total number of trials is known beforehand.
 - (The distribution of a count depends on how the data are produced. The binomial setting is a common situation.)

Binomial Distribution

- What does a binomial distribution look like?
- It depends
 - on two parameters
 - Probability of Success (P)
 - Sample Size/Number of Trials (N)
- (Recall that the normal distribution depends on μ and σ)

Binomial Requirements

1. There are a fixed number, n , of observations.
2. The n observations are all independent. That is, knowing the result of one observation does not change the probabilities we assign to other observations.
3. Each observation falls into one of just two categories, which, for convenience, we call “success” and “failure.”
4. The probability of a success, call it p , is the same for each observation.

Formula for Binomial Probabilities

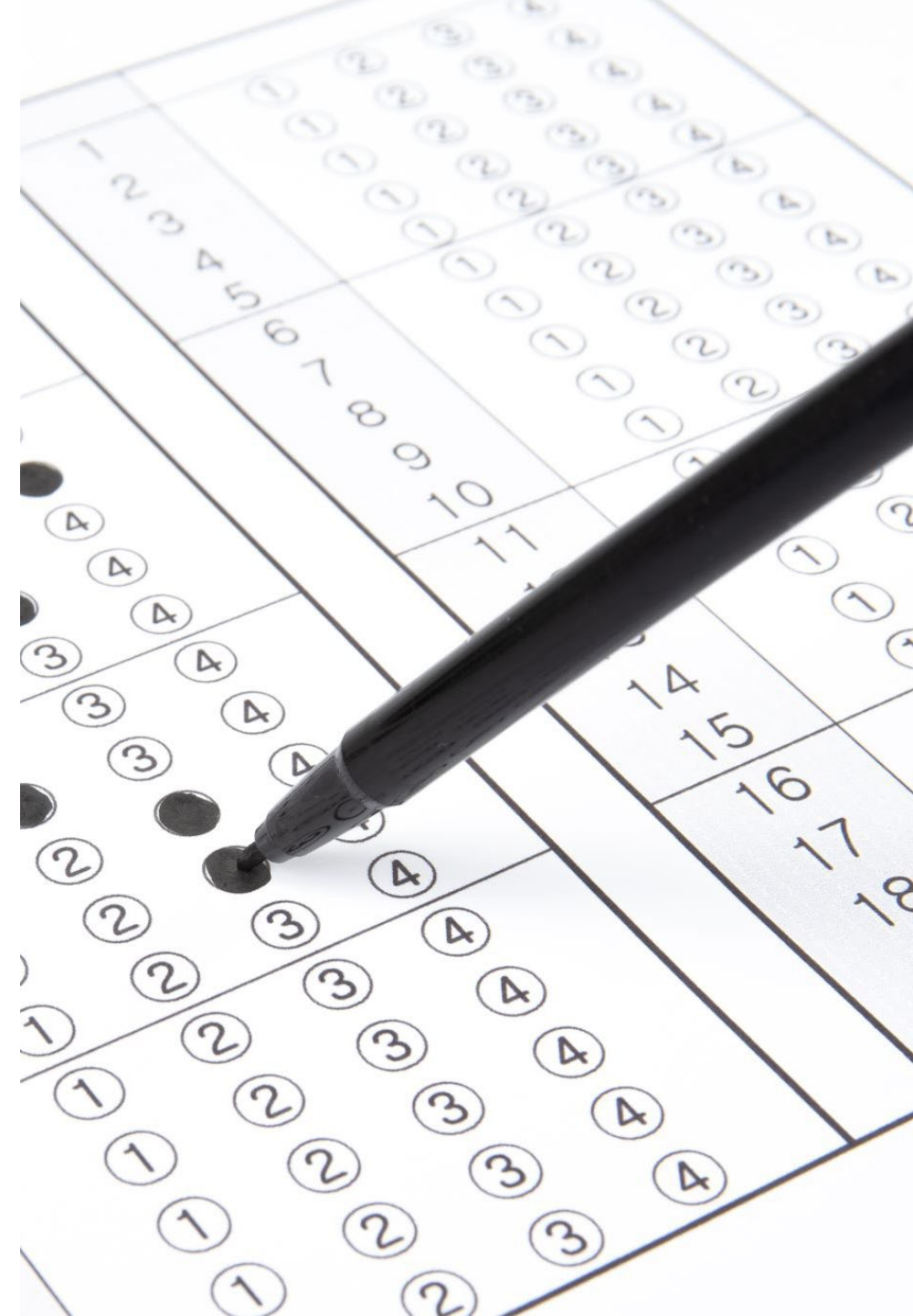
- $P(k) = \frac{N!}{k!(N-k)!} p^k (1-p)^{N-k}$
- where, $P(k)$ is the probability of k successes out of N trials, N is the number of trials and p is the probability of success
- Sometimes this equation is written as
- $P(X=k | pN) = \binom{n}{k} p^k (1-p)^{N-k}$
 - Reads as n chose k

- Chose K numbers from N options $\binom{n}{k}$
- $\frac{N!}{k!(N-k)!}$
- Note: $0! = 1$
- $K! = K(K-1)(K-2)(K-3) \dots 3 \dots 2 \dots 1$
- $6!/4! = 6 \cdot 5$

N chose K

Sampling Distribution of a Count

Choose an SRS of size n from a population with proportion p of successes. When the sample size is less than 5% of the population size, the count X of successes in the sample has approximately the binomial distribution with parameters n and p .



Binomial mean and standard deviation

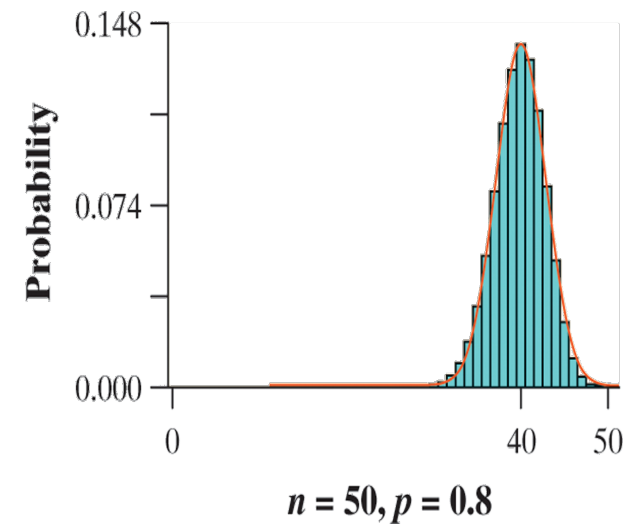
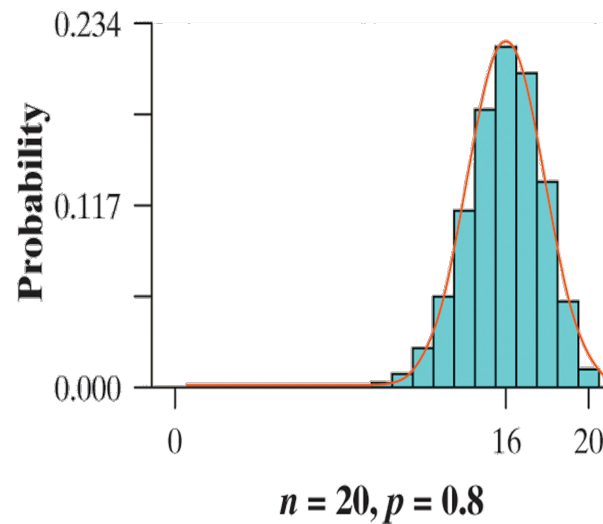
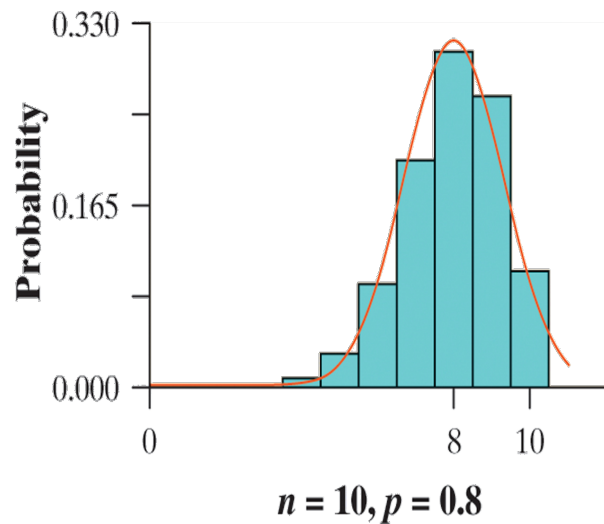
- If a count X has the binomial distribution based on n observations with probability p of success, what is its mean, μ ?
- If a count X has the binomial distribution with number of trials n and probability of success p , the mean and standard deviation of X are:

$$\mu = np$$

$$\sigma = \sqrt{np(1-p)}$$

Normal Approximation

- As n gets larger, something interesting happens to the shape of a binomial distribution.



Normal Approximation

- Suppose that a count X has the binomial distribution with n observations and success probability p .
- When n is large, the distribution of X is approximately Normal,
 - $N(np, \sqrt{np(1-p)})$.
- As a rule of thumb, we will use the Normal approximation when n is so large that $np \geq 10$ and $n(1-p) \geq 10$.



Sampling Distribution

- Simple Idea: Distribution of a statistic
 - Compare to the distribution of a given variable
- Formal Definition:
 - sampling distribution is the distribution of values taken by a statistic across all possible samples of a given size, drawn from a particular population

All statistics have a sampling distribution

- $\bar{x}$, mode, third quartile, upper extreme, etc
- Terminology:
 - $\mu_x \sigma_x$
 - Raw score distribution
 - $\mu_{\bar{x}} \sigma_{\bar{x}} \mu_{\bar{y}}$
 - Parameters of the sampling distribution of the particular statistic

Population Distributions are different

Be careful: The population distribution describes the individuals that make up the population.

A sampling distribution describes how a statistic varies in many samples from the population.

General Equation

- In general,

- $\mu_{\bar{x}} = \mu_x$
- $\sigma_{\bar{x}} = \frac{\sigma_x}{\sqrt{N}}$
- $\sigma_{\bar{x}}^2 = \frac{\sigma_x^2}{N}$

- Because the mean of the statistic $\bar{x}$ is always equal to the mean μ of the population
 - (that is, the sampling distribution of $\bar{x}$ is centered at μ),
 - we say the statistic $\bar{x}$ is an unbiased estimator of the parameter μ .
 - Note: on any particular sample, $\bar{x}$ may fall above or below μ .

Standard Deviation of the Sampling Distribution

- Because the standard deviation of the sampling distribution of $\bar{x}$ is $\sigma/\sqrt{n}$,
 - the averages are less variable than individual observations, and
 - averages are less variable than the results of small samples.
- Not only is the standard deviation of the distribution of $\bar{x}$ smaller than the standard deviation of individual observations,
 - it gets smaller as we take larger samples.
 - The results of large samples are less variable than the results of small samples.
- Although the standard deviation of the distribution of $\bar{x}$ gets smaller, it does so at the rate of $\sqrt{n}$, not n .
 - To cut the sampling distribution's standard deviation in half, for instance,
 - you must take a sample four times as large, not just twice as large.



The Law of Large Numbers

- Although a single statistic, like $\bar{x}$, is rarely identical to the population mean, we can still use it as a reasonable estimate of the population mean.
- Why?
- If we keep taking larger and larger samples, the statistic $\bar{x}$ is guaranteed to get closer and closer to the parameter μ .

LAW OF LARGE NUMBERS

- Draw observations at random from any population with finite mean μ . As the number of observations drawn increases, the mean $\bar{x}$ of the observed values tends to get closer and closer to the mean μ of the population.

Central Limit Theorem

- Similarly, the Central Limit Theorem describes the shape of a sampling distribution.
- Draw an SRS of size n from any population with mean μ and finite standard deviation σ .
 - The central limit theorem says that when n is large, the sampling distribution of the sample mean $\bar{x}$ is approximately Normal:
 - $\bar{x}$ is approximately $N\left(\mu, \sigma/\sqrt{n}\right)$
- No matter what the shape of the raw score distribution.
- The central limit theorem allows us to use Normal probability calculations to answer questions about sample means from many observations,
 - even when the population distribution is not Normal (e.g, skewed, uniform, bimodal)

Hypothesis Testing

What is hypothesis testing?

- A statistical hypothesis is an assumption about a population parameter.
 - This assumption may or may not be true.
- Hypothesis testing refers to the formal procedures used by statisticians to accept or reject statistical hypotheses.
- Sometimes this is called: significance testing
 - A test of significance is a formal procedure for
 - comparing observed data with a claim
 - (also called a hypothesis) whose truth
 - we want to assess.

Null Hypothesis

- The null hypothesis refers to a general default – no relationship between two measured phenomena or a potential medical treatment has no effect.
- Rejecting the null or disproving the null is a central task in modern practice of science
- ****Note**** The claim about the population that we are trying to find evidence *for* is the **alternative hypothesis**.

Juris prudence model of statistical inference

- The Jury
 - skeptical scientific community
- The Prosecutor
 - researcher
- The Defendant
 - Null Hypothesis
- The Detective
 - Statistician



Definitions

- **effect size**, or the salience of the treatment relative to the noise in measurement
- **alpha level** (α , or significance level), or the odds that the observed result is due to chance
- **power**, or the odds that you will observe a treatment effect when it occurs

Type I Error



Wrongly Reject the Null



The hypothesis of innocence is only rejected when an error is very unlikely, because one doesn't want to convict an innocent defendant.



Such an error is called *error of the first kind* (i.e., the conviction of an innocent person), and the



occurrence of this error is controlled to be rare.

Set this value to 5%

Type II Error

- Wrongly Accept the Null
- If the null is not true, but the data say it is – then we are wrongly accepting the Null.
- In STEM, we prioritize reducing the FALSE Positives.
- As a consequence of this,
 - the error of the second kind (acquitting a person who committed the crime),
 - is often rather large.
 - In psychology this is commonly 20%.

Power and Type Two Error

- We cannot control the rate of Type II errors directly.
- Instead, we can influence our ability to detect an effect, given that the null is false
 - Power = 1 – Type II Error
 - In psychology, we aim for 80%.

Phenomenal cosmic powers...

Ways to influence power

- Sample Size
- Standard Deviation
- Significance Level
 - Type 1 Error
- Strength of our effect



Null Hypothesis

- The null hypothesis refers to a general default – no relationship between two measured phenomena or a potential medical treatment has no effect.
- Rejecting the null or disproving the null is a central task in modern practice of science
- First use – *The Design of Experiments* (1935) by Ronald Fisher
 - "never proved or established, but is possibly disproved, in the course of experimentation".

Quantifying the Unusual

One way of quantifying how unusual a result is involves a probability calculation.

A result X is unusual if the probability of a result more extreme than X is small.

Quantifying the Unusual



We are trying to show a group is unusual



In standard hypothesis testing, we employ a strategy that is based on falsifying the opposite of what we are trying to show.

That is, we begin by stating a hypothesis (null hypothesis) that is the opposite of what we believe: the group is normal

What we then seek to do is show that, in fact, the group is unusual,

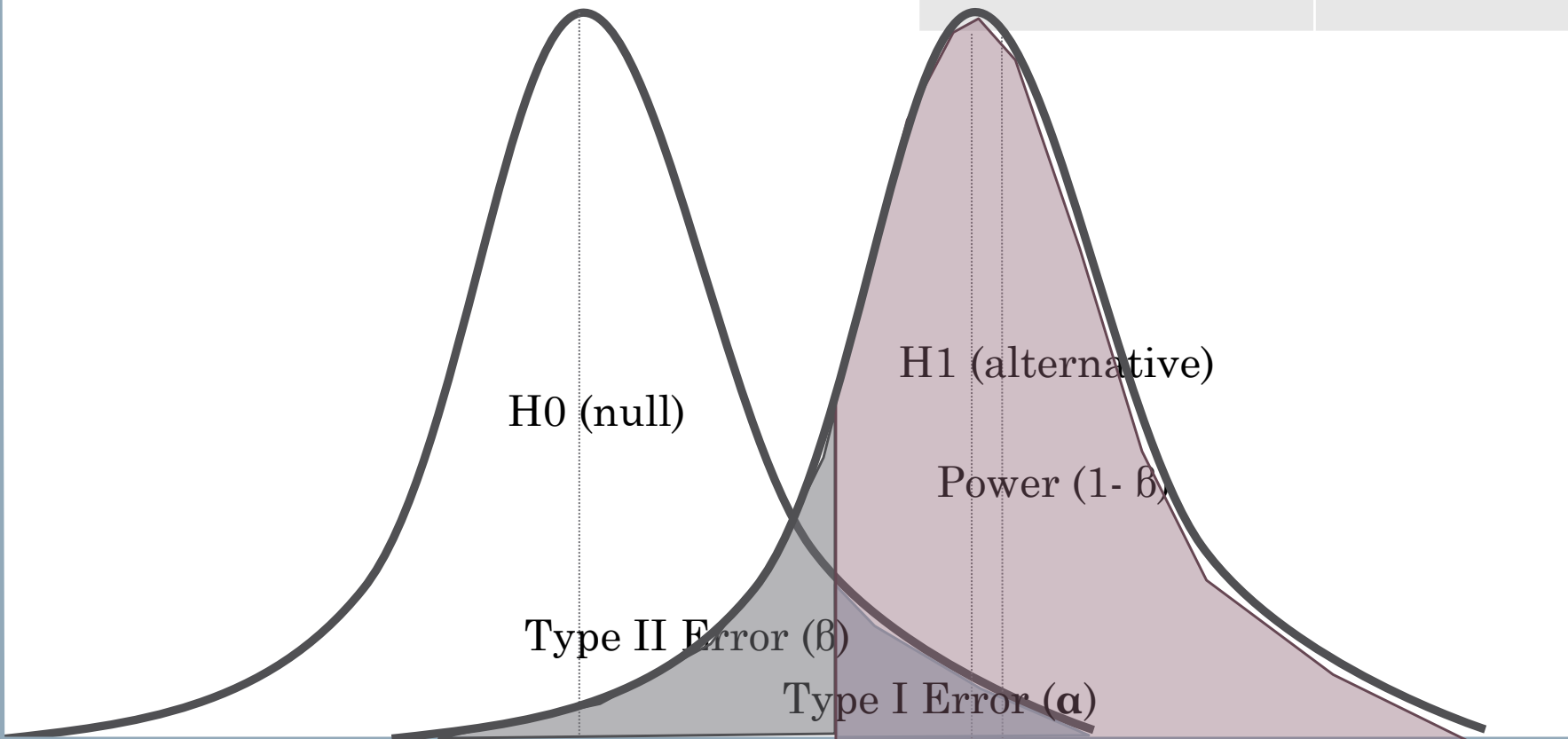
- thereby falsifying the opposite of what we wish to show, and
- demonstrating the truth of what we wish to show.

Quantifying the Unusual

- How can we falsify what we don't believe (that is, the null hypothesis)?
 - We need a way of demonstrating that an observation (our data) is not frequent enough to be unlikely to have occurred by chance.
 - By demonstrating that, we can reject the null hypothesis (that the group is normal), and accept the alternative hypothesis (that the group is unusual)

H0: the group is not unusual
H1: The group is unusual

	H ₀ is true not unusual	H ₁ is true unusual
Accept null hypothesis	Right decision	Wrong decision Type II Error (β)
Reject null hypothesis	Wrong decision Type I Error (α)	Right decision (1- β)



Calculating p Values



The preceding considerations have led to some standard conventions for calculating p values (or “significance levels”) in the continuous case.



In statistical hypothesis testing, it is common to test a hypothesis about a parameter with a “test statistic” that is some function of an estimate of that parameter.

Calculating p Values

- For example, we test a hypothesis that a population mean μ is equal to 100 by examining a test statistic t that is a function of an estimate of how much μ deviates from 100.
- If the statistical null hypothesis is true, then the test statistic has a known distribution,
 - the “null sampling distribution.”

Calculating p Values

- If a value occurs that is unlikely to occur in that distribution, we consider the result as evidence against the null.
- In one-sided (or one-tailed) testing situations,
 - evidence to reject the null can come at only one end of the number line, while
- in two-sided (or two-tailed) situations, the evidence can come at either end

Calculating p Values



We determine whether our statistical result is significant or not by looking at the p value.



In general, the lower the p-value, the “more significant” the result.

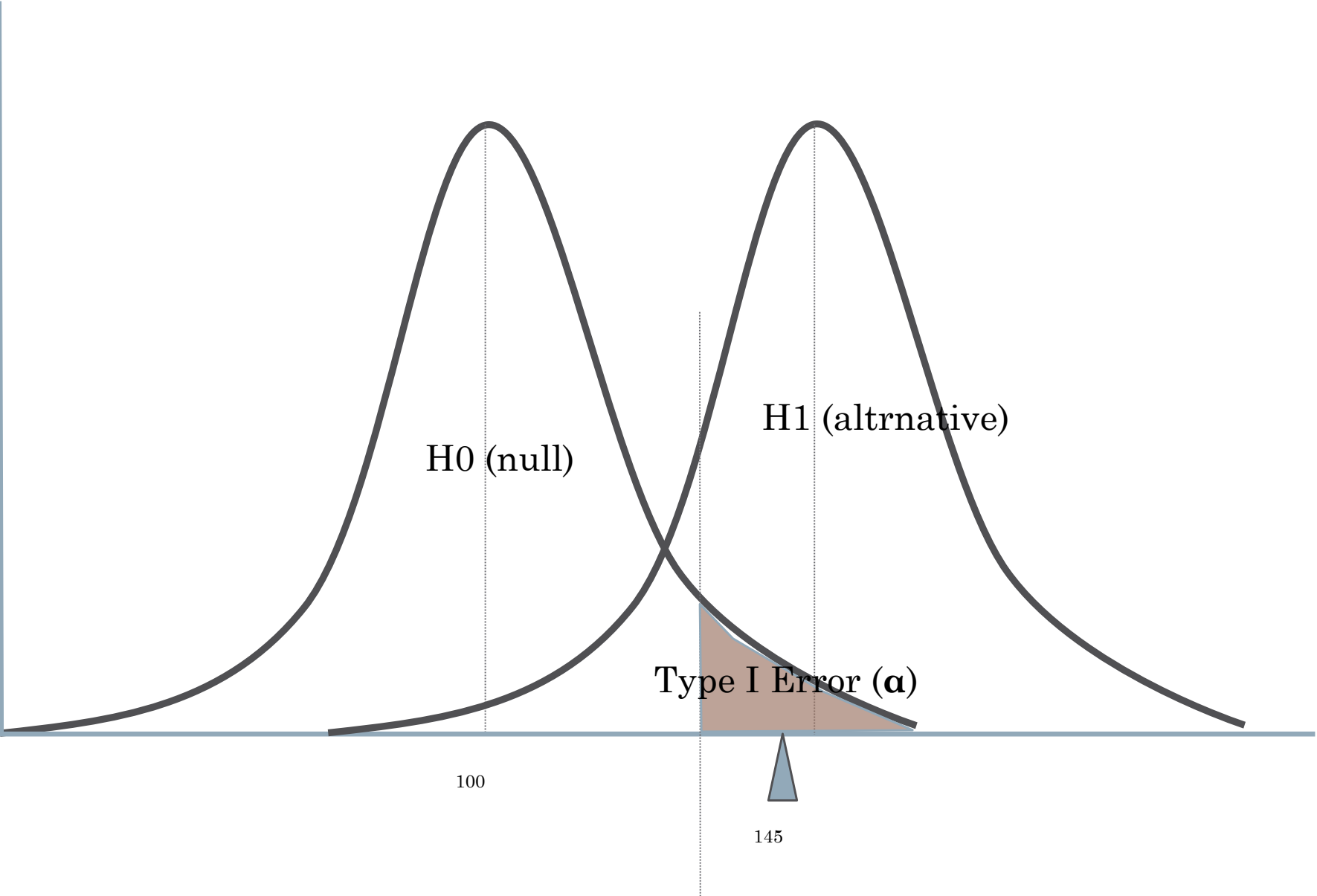


In particular, if the p-value is less than α , we say that the result is “significant at the α level.”

Calculating p Values

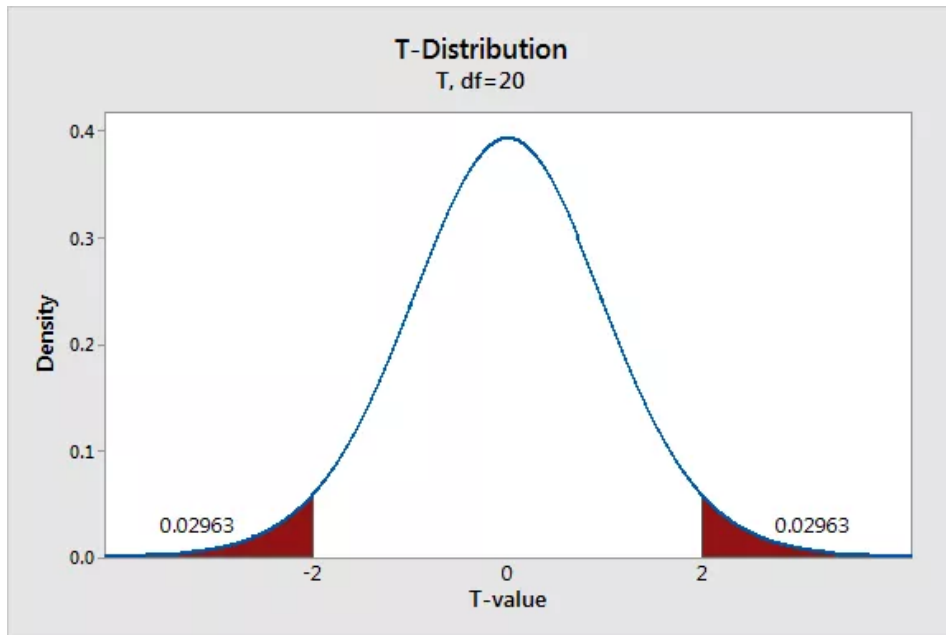
- The p-value is the probability of getting a result as extreme or more extreme as the observed value:
- 1) If the test is one-sided,
 - the calculation is performed in the direction that
 - would result in rejection, and
 - the p value is the probability of
 - obtaining a result more extreme than the
 - observed statistic in that rejection direction.

density



Calculating p Values

- 2) If the test is two-sided, the calculation is performed toward the nearest rejection side, then the resulting probability is doubled.



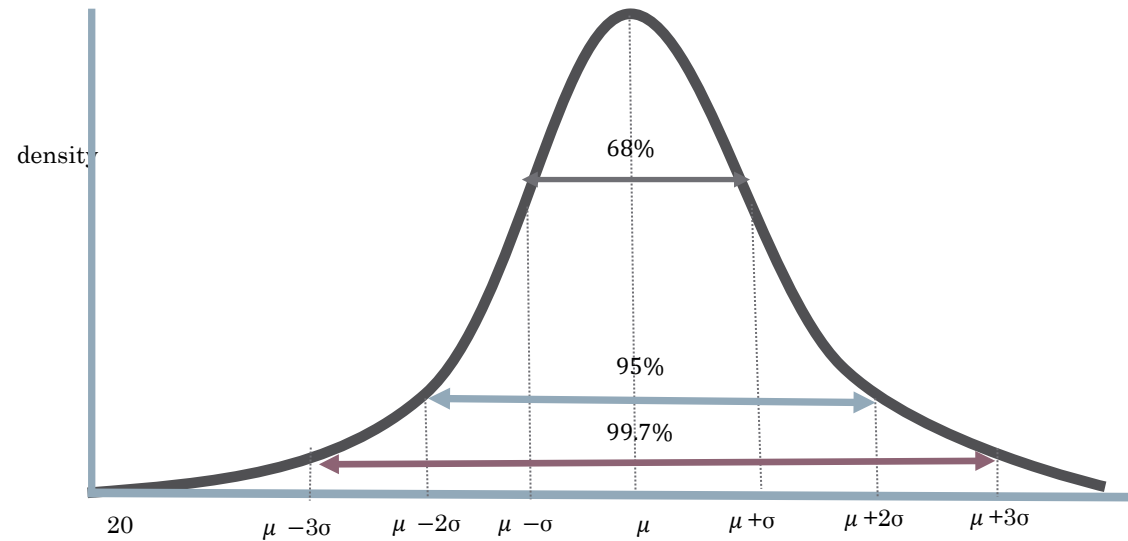
Calculating p Values

- Since the value is lower than 0.05, we reject the null hypothesis at the 0.05 level, but not at the 0.01 level, since the value is larger than 0.01.

T Tests

Review: z-distribution

- A standard normal distribution with a mean of 0 and standard deviation of 1.



Z-score

- According to CLT, z- score is obtained by the following formula

$$z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

- We refer to the denominator here as **standard error**: they tell us how much statistics, such as the mean, vary from one sample to another
 - Large standard error: someone else who replicate your study is very likely to find a sample mean quite different from yours.

	Z-distribution	T-distribution
Do we know population variance σ^2	Yes.	No. We substitute sample variance s^2 for σ^2
shape	Bell shaped and symmetric	Bell shaped and symmetric, but fatter tail than z-distribution.
Mean	0	0
variance	1	Df/(df-2): proportionally larger than z
score	$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$	$t = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}}$

More about t-distribution

- the distribution of t varies as a function of the **degrees of freedom (df)**, which for the moment we will define as one less than the number of observations in the sample.

$$• t_{df} = \frac{z}{\sqrt{\frac{\chi^2_{df}}{df}}}$$

- $Df = n - 1$. (n : sample size)
- t approaches standard normal distribution as df approaches infinity, this happens because s estimate σ more accurately as the sample size increases. So using s in place of σ causes little variation when the sample is large

Critical Value of a t-statistic

- We calculate critical value also to reach a statistical conclusion
- Comparing critical value to t statistic is same with comparing p value to α
- **If $p < \alpha$, we reject the null**
- **Similarly, if the result of the t-statistic is larger than the critical value, we reject the null.**

Finding critical value and CI

- We find the critical value in t-table using the degree of freedom and α level.
- Suppose $\alpha=0.05$ for the mean μ of a normal population based on a sample of size $n=12$. what critical value should you use if we are to do a two-sided test?

Upper tail probability p

df	0.05	0.025	0.02	0.01
10	1.812	2.228	2.359	2.764
11	1.796	2.201	2.328	2.718
12	1.782	2.179	2.303	2.681
Z*	1.645	1.960	2.054	2.326
CI	90%	95%	96%	98%

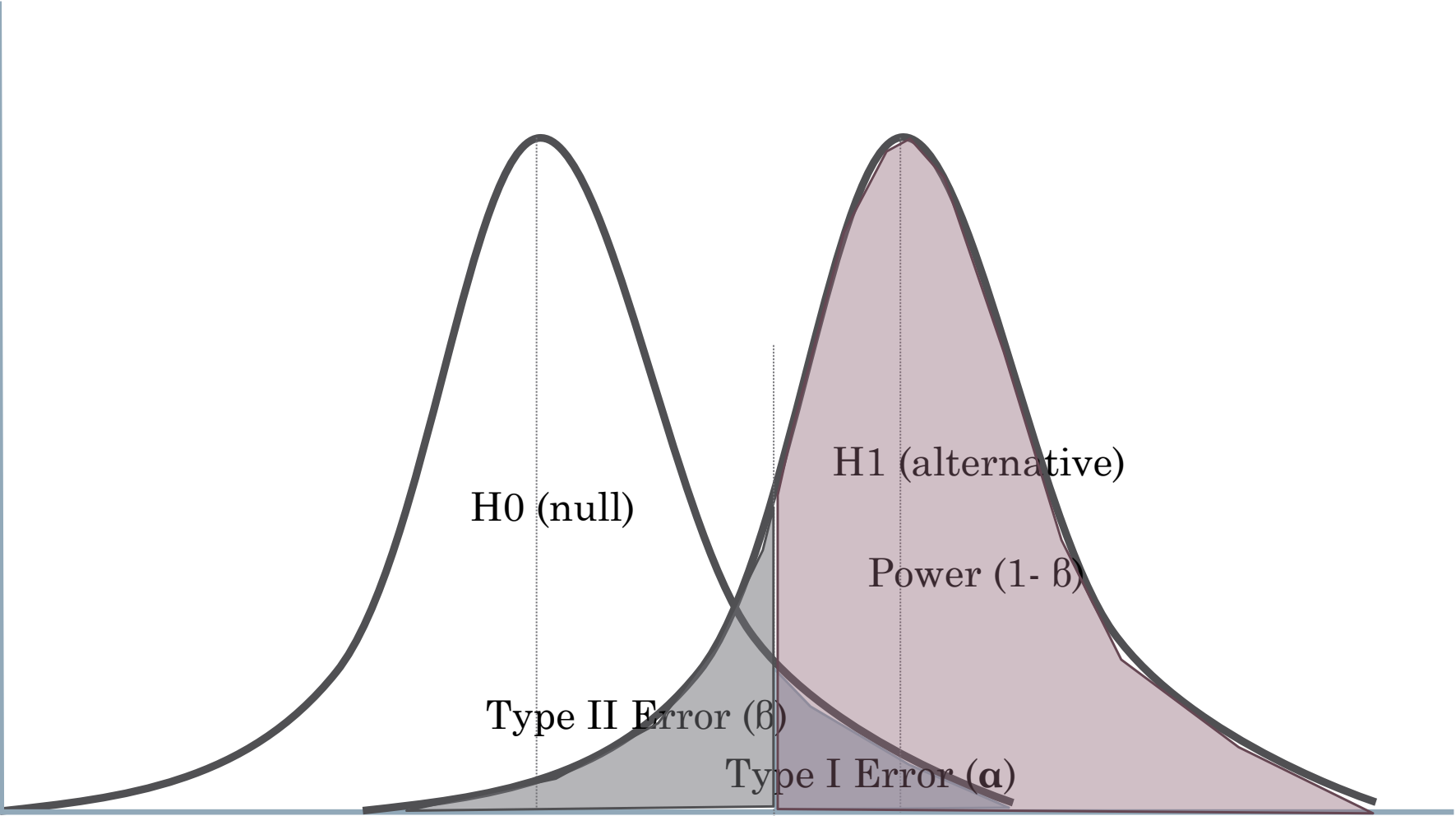
Confidence Interval: we'd like to retain

- CI: $\bar{x} \pm t^* \frac{s}{\sqrt{n}}$
- t^* is the critical value we found in last step.
- $\bar{x}$: sample mean
- s : sample standard deviation
- n : sample size

Types of t-test

- One sample t-test
 - Used when we want to know whether a sample we collected come from a particular population with unknown mean μ . (similar to what we did with z-test so far)
- Matched pair t-test
 - Used when the two samples of data were related or provided by the same participants (i.e., pre and post test)
- Independent sample t-test
 - test the difference between the means of two independent groups (i.e., treatment and control group)

density



7 Steps

1. **Decide what type of test we want to use**

1. We don't know population sd $\rightarrow$ t-test
2. We only have one sample and we want to know whether it is from a particular population $\rightarrow$ one sample t-test

2. **decide what the null and alternative hypothesis is.**

- Null: students are guessing. $H_0: \mu = 20$ (μ_0)
- Alternative: students are not guessing. $H_a: \mu \neq 20$

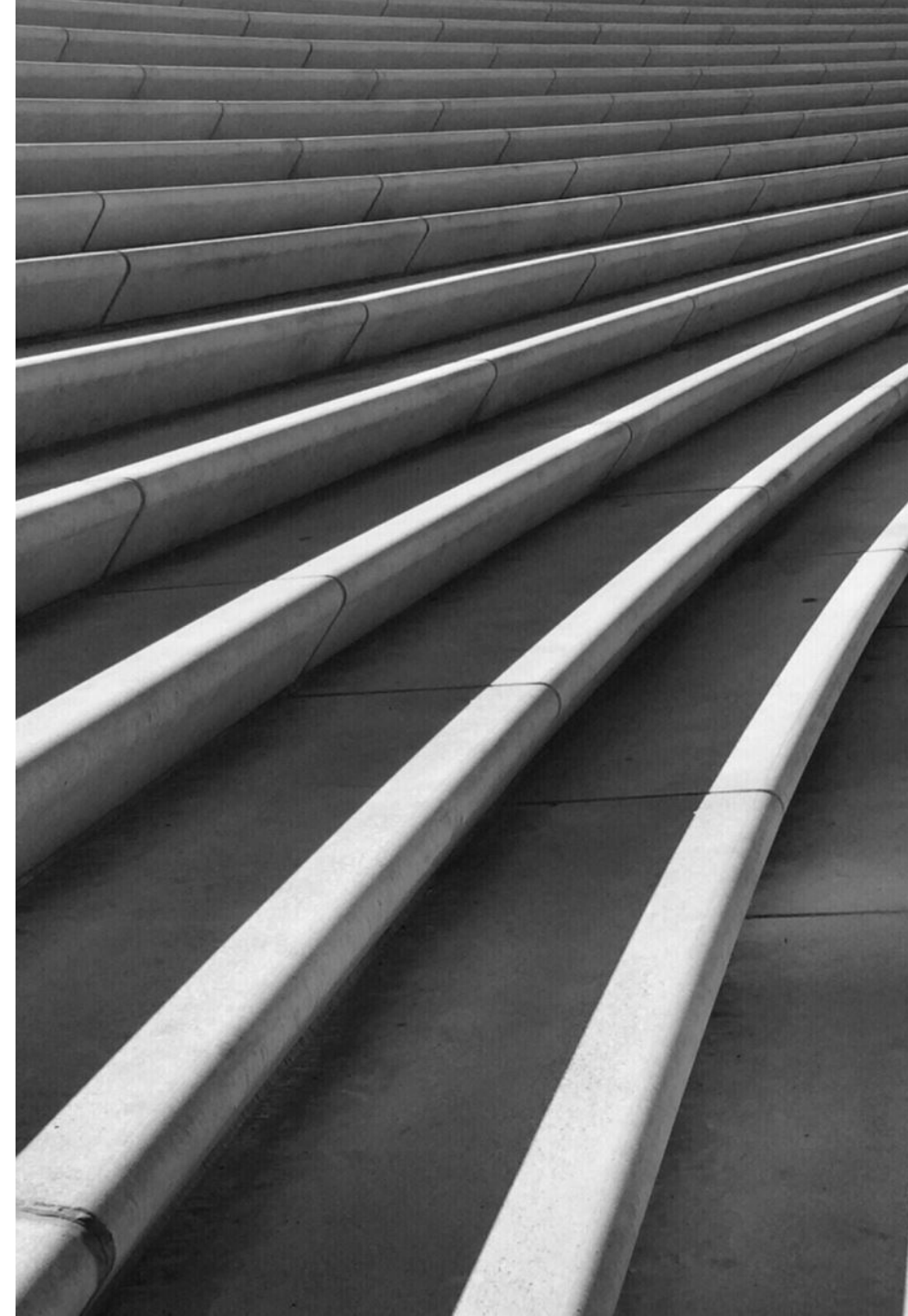
Steps (continued)

3. List what we have

- $\bar{x} = 46.21$
- $\mu = 20$
- $N = 20$
- $s = 6.73$

4. Compute t-statistics

$$t = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}} = \frac{46.21 - 20}{\frac{6.73}{\sqrt{28}}} = 20.61$$



Steps (continued)

5. finding critical value in t-table

- $Df = n - 1 = 27$
- Since we specified the test as two-tailed, a 95% CI will have a upper tail probability of $0.05/2 = 0.025$
- $t^* = 2.051$

6. Compare t-statistic to t^* (we can also calculate p and compare to α)

- $20.61 > 2.051$

7. Make decision: reject null or not, and draw conclusion

- We reject null hypothesis based on the results, the group of students are not guessing.

Matched pair t-test

- Used when we want to know whether a sample we collected come from a particular population with unknown mean μ .
- Important: we are only interest in the difference score, and we don't make inference about pre-test measure or post-test measure
 - We only use the difference score to denote change, thus a matched pair t-test is same with the one-sample t-test in terms of procedure.
- In one sample t-test, we have the sampling distribution of the mean, here we instead have the sampling distribution of the difference score



Independent sample t-test

- Hypothetical set of means and differences between means when sampling from two populations

sample means (random draw)	$\bar{X}_{11}$ $\bar{X}_{12}$ $\bar{X}_{13}$... $\bar{X}_{1\alpha}$	$\bar{X}_{21}$ $\bar{X}_{22}$ $\bar{X}_{23}$... $\bar{X}_{2\alpha}$	$\bar{X}_{11} - \bar{X}_{21}$ $\bar{X}_{12} - \bar{X}_{22}$ $\bar{X}_{13} - \bar{X}_{23}$... $\bar{X}_{1\alpha} - \bar{X}_{2\alpha}$
MEAN	μ_1	μ_2	$\mu_1 - \mu_2$
VARIANCE	$\frac{\sigma^2_1}{n_1}$	$\frac{\sigma^2_2}{n_2}$	$\frac{\sigma^2_1}{n_1} + \frac{\sigma^2_2}{n_2}$
S.D.	$\frac{\sigma_1}{\sqrt{n_1}}$	$\frac{\sigma_2}{\sqrt{n_2}}$	$\sqrt{\frac{\sigma^2_1}{n_1} + \frac{\sigma^2_2}{n_2}}$

Upper part:
Individual sample means in a sampling distribution

Lower part:
properties of the sampling distribution

MEAN	μ_1	μ_2	$\mu_1 - \mu_2$
VARIANCE	$\frac{\sigma^2_1}{n_1}$	$\frac{\sigma^2_2}{n_2}$	$\frac{\sigma^2_1}{n_2} + \frac{\sigma^2_2}{n_2}$
S.D.	$\frac{\sigma_1}{\sqrt{n_1}}$	$\frac{\sigma_2}{\sqrt{n_2}}$	$\sqrt{\frac{\sigma^2_1}{n_2} + \frac{\sigma^2_2}{n_2}}$

1. we are concerned with testing differences between means. The mean of this distribution is equal $\mu_1 - \mu_2$.

MEAN	μ_1	μ_2	$\mu_1 - \mu_2$
VARIANCE	$\frac{\sigma^2_1}{n_1}$	$\frac{\sigma^2_2}{n_2}$	$\frac{\sigma^2_1}{n_2} + \frac{\sigma^2_2}{n_2}$
S.D.	$\frac{\sigma_1}{\sqrt{n_1}}$	$\frac{\sigma_2}{\sqrt{n_2}}$	$\sqrt{\frac{\sigma^2_1}{n_2} + \frac{\sigma^2_2}{n_2}}$

2. The variance of this distribution is given by what is commonly called the **Variance Sum Law**

- **Variance Sum Law states that** the variance of the sum or difference of two *independent* variables is equal to the sum of their variances.
- $\sigma^2_{\bar{X}_1 - \bar{X}_2} = \sigma^2_{\bar{X}_1} + \sigma^2_{\bar{X}_2} = \frac{\sigma^2_1}{n_1} + \frac{\sigma^2_2}{n_2}$

MEAN	μ_1	μ_2	$\mu_1 - \mu_2$
VARIANCE	$\frac{\sigma^2_1}{n_1}$	$\frac{\sigma^2_2}{n_2}$	$\frac{\sigma^2_1}{n_2} + \frac{\sigma^2_2}{n_2}$
S.D.	$\frac{\sigma_1}{\sqrt{n_1}}$	$\frac{\sigma_2}{\sqrt{n_2}}$	$\sqrt{\frac{\sigma^2_1}{n_2} + \frac{\sigma^2_2}{n_2}}$

3. Shape of the third sampling distribution?

Rule: the sum or difference of two independent normally distributed variables is itself normally distributed.

- because we know the sampling distribution of means is at least approximately normal for reasonable sample sizes
- then the distribution in third column must itself be at least approximately normal.

t-statistic for two independent sample

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

1. Because the null hypothesis is generally the hypothesis that $\mu_1 - \mu_2 = 0$, we usually drop that term from the equation
2. Df of a two sample t-test: $(n_1-1)+(n_2-1)=n_1+n_2-2$
3. CI: $(\bar{x}_1 - \bar{x}_2) \pm t^* \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$

Robustness of t-test

A confidence interval or significance test is called robust if the confidence level or p -value does not change very much when the conditions for use of the procedure are violated.



Rule of thumb

For one sample t-test or matched pair t-test

- For $n < 15$, use t only if the data appear close to Normal or roughly symmetric. If the data are clearly skewed or if outliers are present, do not use t .
- For $n > 40$: The t procedures can be used, even for clearly skewed distribution

Robustness of t-test

- For two independent t-test
 - More robust than the one-sample t methods, particularly when the distributions are not symmetric.
 - **When $n_1=n_2$ and the two population have same shape**, t is robust for even very small sample. i.e., $n_1=n_2=5$.
 - Rule of thumb: adapt the guidelines for one-sample t procedures to two-sample procedures by replacing “sample size” with the “sum of the sample sizes,” $n_1 + n_2$.

Comparing More than Two Groups

ANOVA, Regression, etc

analysis of variance (ANOVA)

- Is method for comparing any number of groups.
- It can be thought of as an extended multi-group t-test
 - OR as a regression with grouping variables
- Is an efficient alternative to running n t-tests to compare n-groups

Why can't we Just Run a Bunch of t-tests?

- For example, in a three group design, we could just run 3 t-tests?
- Problems:
 - Many t-tests with lots of groups
 - Alpha-control
 - Increase type-1 error rate (because we're testing multiple times)
 - Multiplication of type 1 errors
 - e.g, If we test 20 times, we'll find 1 significant result by chance (if $\alpha = .05$)
 - <http://shinyapps.org/apps/p-hacker/>
 - ANOVA controls alpha =.05 across all tests (as does regression)

Consequences of Inflated Type 1 Errors

- False Positives (Failure to Replicate)
 - Remember that a p-value indicates the probability that the result occurred under the null hypothesis.
- 'Replication Crisis'

'Replication Crisis'

- Scientists have found that the results of many scientific studies are
 - difficult or impossible to replicate on subsequent investigation,
 - either by independent researchers or
 - by the original researchers themselves.
- In the early 2010s, as part of a growing awareness of the problem, phenomenon was deemed a "crisis"

Attempt to Replicate Psychology

- A report by the Open Science Collaboration in August 2015
 - estimated the reproducibility of 100 studies in psychological science from three high-ranking psychology journals.
- Overall, 36% of the replications yielded significant findings (p value below .05)
 - compared to 97% of the original studies that had significant effects.
- The mean effect size in the replications was approximately half the magnitude of the effects reported in the original studies.

Issue is broader than psychology

2016 poll of 1,500 scientists reported in *Nature*,

70% of them had failed to replicate at least one other scientist's experiment (50% had failed to replicate one of their own experiments).

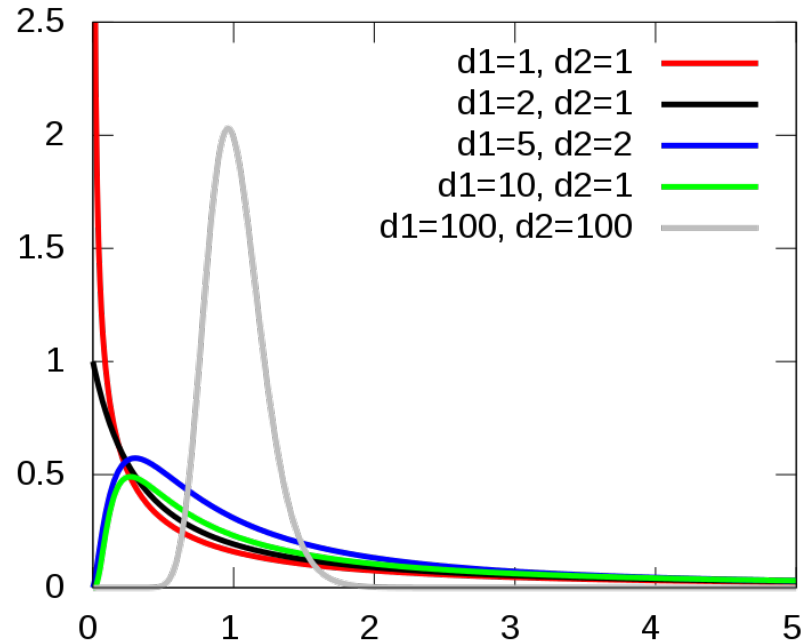
Results differed by field

- chemistry: 90% (60%),
- biology: 80% (60%),
- physics and engineering: 70% (50%),
- medicine: 70% (60%),
- earth and environment science: 60% (40%).

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$$f(x; d_1, d_2) = \frac{\sqrt{\frac{(d_1 x)^{d_1} d_2^{d_2}}{(d_1 x + d_2)^{d_1 + d_2}}}}{x B\left(\frac{d_1}{2}, \frac{d_2}{2}\right)}$$

$$= \frac{1}{B\left(\frac{d_1}{2}, \frac{d_2}{2}\right)} \left(\frac{d_1}{d_2}\right)^{\frac{d_1}{2}} x^{\frac{d_1}{2} - 1} \left(1 + \frac{d_1}{d_2} x\right)^{-\frac{d_1 + d_2}{2}}$$

Equation is for the PDF. Note: B is the beta function

F Distribution

- Looks like a skewed normal for many combinations of parameters
- Has two parameters
 - Degrees of freedom in the numerator
 - Degrees of Freedom in the denominator
- Critical Value is in the tail
 - All ANOVAs are one-tailed tests
 - In the Upper Tail

Multiple comparisons

- Statistical methods for dealing with multiple comparisons usually have two steps:
 1. An *overall test* to see if there is good evidence of any differences among the parameters that we want to compare.
 2. A detailed *follow-up analysis* to decide which of the parameters differ and to estimate how large the differences are.
- The overall test, though more complex than the tests we have met to this point, is reasonably straightforward. Formal follow-up analysis can be quite elaborate. We will concentrate on the overall test and use data analysis to describe in detail the nature of the differences.

The analysis of variance F test

- We want to test the null hypothesis that there are no differences among the means of the populations.

$$H_0: \mu_1 = \mu_2 = \mu_3$$

- The basic conditions for inference are that we have random samples from the three populations and that flower lengths (in our example) are Normally distributed in each population.
- The alternative hypothesis is that there is some difference. That is, not all means are equal.

$$H_a: \text{not all of } \mu_1, \mu_2, \text{ and } \mu_3 \text{ are equal}$$

The analysis of variance F test

- This alternative hypothesis is not one-sided or two-sided. It is “many-sided.”
- This test is called the analysis of variance (ANOVA) F test.

EXAMPLE: Comparing tropical flowers: ANOVA (See `anovaexample.R`)

- **SOLVE** (inference): Software tells us that, for the flower length data in the test statistic is $F = 259.12$ with $P < 0.0001$. There is very strong evidence that the three varieties of flowers do not have the same mean length. It appears from our preliminary data analysis that *bihai* flowers are distinctly longer than either red or yellow. Red and yellow are closer together, but the red flowers tend to be longer.
- **CONCLUDE**: There is strong evidence ($P < 0.0001$) that the population means are not all equal. The most important difference among the means is that the *bihai* variety has longer flowers than the red and yellow varieties.



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The ANOVA F statistic

- The analysis of variance F statistic for testing the equality of several means has this form:

$$F = \frac{\text{variation between the sample means}}{\text{variation among individuals in the same sample}}$$

- F must be zero or positive, with F being zero only when all sample means are identical and F getting larger as means move further apart.
- Large values of F are evidence against H_0 : equal means; while the alternative H_a is many-sided, the F test is upper one-sided.



Some details of ANOVA

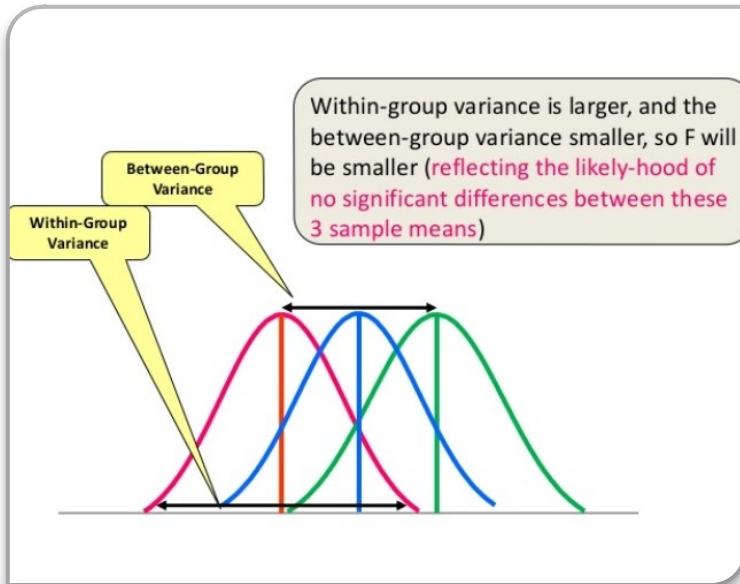
- The mean square in the numerator of F is an average of the squares of these deviations. We call it the mean square for groups (MSG):

- $$\text{MSG} = \frac{n_1(\bar{x}_1 - \bar{x})^2 + n_2(\bar{x}_2 - \bar{x})^2 + \dots + n_I(\bar{x}_I - \bar{x})^2}{I - 1}$$

- The mean square in the denominator of F measures variation among individual observations in the same sample.
- For all I samples together, we use an average of the individual sample variances. We call it the mean square for error (MSE):

- $$\text{MSE} = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2 + \dots + (n_I - 1)s_I^2}{N - I}$$

Some details of ANOVA



THE ANOVA F TEST

- Draw an independent SRS from each of I Normal populations that have a common standard deviation but may have different means. The sample from the i^{th} population has size n_i , sample mean $\bar{x}_i$, and sample standard deviation s_i .
- To test the null hypothesis that all I populations have the same mean against the alternative hypothesis that not all the means are equal, calculate the **ANOVA F statistic**:

$$F = \frac{\text{MSG}}{\text{MSE}}$$



Some details of ANOVA

- The numerator of F is the **mean square for groups**:

$$\text{MSG} = \frac{n_1(\bar{x}_1 - \bar{x})^2 + n_2(\bar{x}_2 - \bar{x})^2 + \cdots + n_I(\bar{x}_I - \bar{x})^2}{I - 1}$$

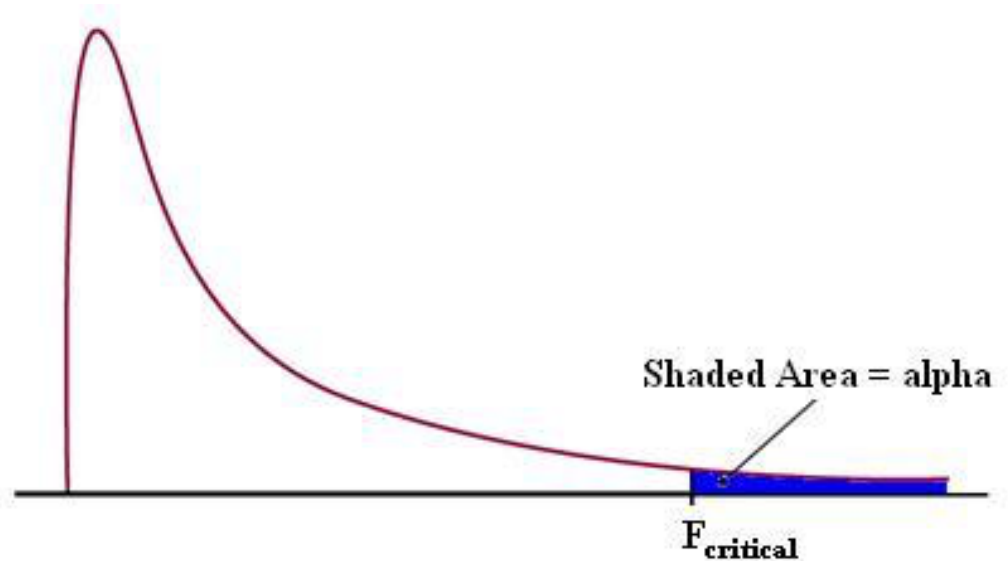
- The denominator of F is the **mean square for error**:

$$\text{MSE} = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2 + \cdots + (n_I - 1)s_I^2}{N - I}$$

- When H_0 is true, F has the **F distribution** with $I - 1$ and $N - I$ degrees of freedom.

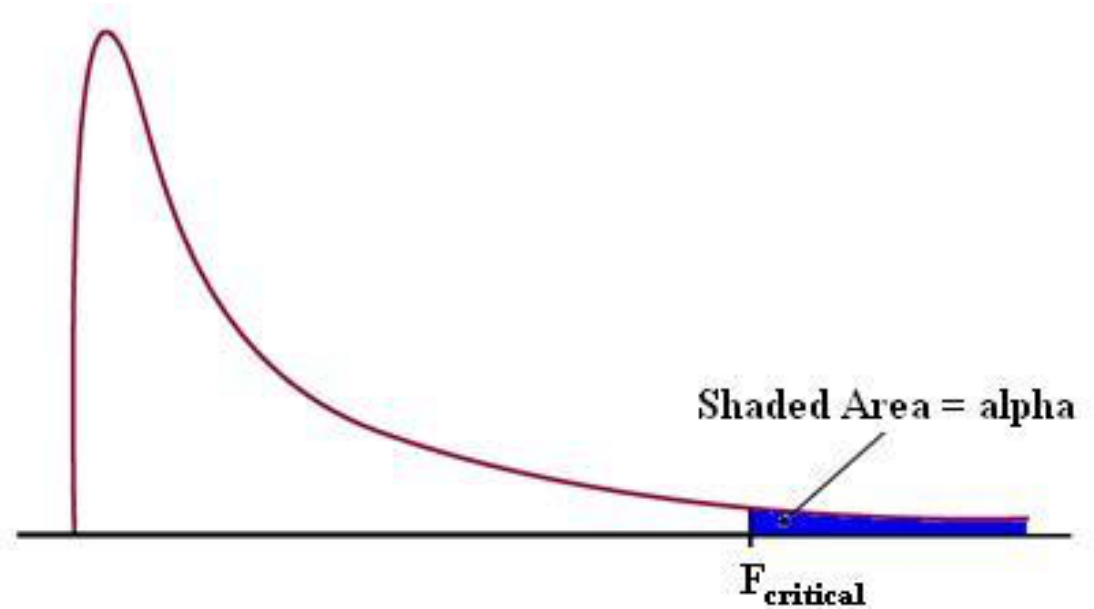
ANOVA and the F statistic

- Therefore, there is only one critical region, in the right tail (shown as the blue shaded region above).
- If the F-statistic lands in the critical region, we can conclude that the means are significantly different and we reject the null hypothesis.



ANOVA and the F statistic

- Again, we have to find the critical value to determine the cut-off for the critical region. We'll use the F-table for this purpose.
- We need to look at different F-values for each alpha/significance level because the F-critical value is a function of two things: df_1 and df_2



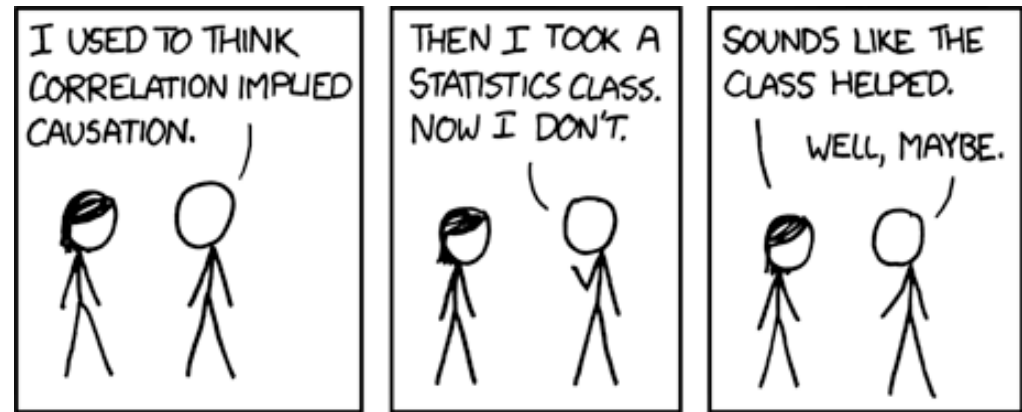
Correlation $\nrightarrow$ Causation



- We cannot assume that correlation means causation
 - A causes B; (direct causation)
 - B causes A; (reverse causation)
 - A and B are consequences of a common cause, but do not cause each other;
 - A causes B and B causes A (bidirectional or cyclic causation);
 - A causes C which causes B (indirect causation);
 - There is no connection between A and B; the correlation is a coincidence.

Correlation $\nrightarrow$ Causation

- “Correlation doesn't imply causation, but it does waggle its eyebrows suggestively and gesture furtively while mouthing look over there.”
 - <https://xkcd.com/552/>
- Correlations are necessary for causation, but not sufficient.



Correlation



Causation

- Although correlation isn't sufficient, it is necessary component for causation.
- Hill's Criteria of Causation
 - Austin Bradford Hill, "The Environment and Disease: Association or Causation?," [Proceedings of the Royal Society of Medicine](#), 58 (1965): 295-300.
- Cook and Campbell's Three Criteria (1979)
 - Cook, T. D. and Campbell, D. T.(1979). Quasi-experimentation: Design & analysis issues for field settings. Boston: Houghton Mifflin.

Hill's Criteria (1965)

Strength	A relationship is more likely to be causal if the correlation coefficient is large and statistically significant.
Consistency	A relationship is more likely to be causal if it can be replicated.
Specificity	A relationship is more likely to be causal if there is no other likely explanation.
Temporality	A relationship is more likely to be causal if the effect always occurs after the cause.
Gradient	A relationship is more likely to be causal if a greater exposure to the suspected cause leads to a greater effect.
Plausibility:	A relationship is more likely to be causal if there is a plausible mechanism between the cause and the effect.
Coherence	A relationship is more likely to be causal if it is compatible with related facts and theories
Experiment	A relationship is more likely to be causal if it can be verified experimentally.
Analogy	A relationship is more likely to be causal if there are proven relationships between similar causes

Cook and Campbell (1979)

Covariation:

- Changes in the assumed cause (X) are related to changes in the assumed effect (Y). Changing X results in a predictable change in Y.

Temporal Precedence:

- The assumed cause must occur before the assumed effect.

No Plausible Alternative Explanations:

- The assumed cause must be the only reasonable explanation for changes in the measured assumed effect.

No Plausible Alternative Explanations

The third criterion is the hardest to satisfy

Plausible alternative explanations are “threats to validity”

Goal is to Minimizing Threats to Validity

- Multiple approaches (can be combined)
 - Argument
 - Measurement or Observation
 - Analysis
 - Preventive Action
 - Design

Goal is to Minimizing Threats to Validity

- The five categories listed should not be considered mutually exclusive.
- The inclusion of measurements designed to minimize threats to validity will obviously be related to the design structure and is likely to be a factor in the analysis.
- A good research plan should, where possible. make use of multiple methods for reducing threats.
- In general, reducing a particular threat by design or preventive action will probably be stronger than by using one of the other three approaches.
- The choice of which strategy to use for any particular threat is complex and depends at least on the cost of the strategy and on the potential seriousness of the threat.